

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC/ME/CL/EE)

November -2013

MA 202: Engineering Mathematics-III

Date: 19.11.2013, Tuesday

Time: 01:30 p.m. to 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION-I

Q-1 (a) Express $f(x) = c - x$ when $0 < x < c$ as a half -range Cosine series with period $2c$. (p=c) [04]

(b) Find Laplace transform of function $\left(\cos^3 2t\right)$. [03]

Q-2 (a) Solve the equation $x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ by Ferrari method. [05]

(b) Find the Fourier series of the periodic square wave function $f(x)$

$$f(x) = -k; \quad -\pi < x < 0 \\ = k; \quad 0 < x < \pi \quad \text{and } f(x+2\pi) = f(x). \text{ Hence deduce that } 1 - \frac{1}{3} + \frac{1}{5} - \dots = \frac{\pi}{4}. \quad [05]$$

(c) Using Convolution theorem, obtain the value of $L^{-1}\left(\frac{1}{s(s^2 + 4)}\right)$. [04]

OR

Q-2 (a) Find inverse Laplace transformation of the function (i) $\frac{s+6}{s^2+2s+2}$ (ii) $\log\left(\frac{s+4}{(s-3)}\right)$ [05]

(b) Solve the equation $x^3 - 18x - 35 = 0$ by Cardon's method. [05]

(c) Find the Fourier series to represent $f(x) = x^2 - 2$ when $-2 \leq x \leq 2$. $p = 2l = 4, l = 2$. [04]

Q-3 (a) Solve the equation by the Laplace transforms method

$$y'' + w^2 y = 0; \quad y(0) = 8, y'(0) = 7. \quad [05]$$

(b) Solve the equation $x^3 - 6x^2 + 6x - 5 = 0$ by Cardon's method. [05]

(c) Find Laplace transformation of the function (i) $e^{-2t} \sin^2 t$ (ii) $\frac{\cos 4t - \cos 2t}{t}$. [04]

OR

Q-3

(a) Find the Fourier series of the periodic function $f(x) = \pi + x; \quad -\pi < x < 0$
 $= \pi - x; \quad 0 < x < \pi$. Hence, [05]

$$\text{deduced that } 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}.$$

(b) Solve the equation $x^4 - 12x^3 + 41x^2 - 18x - 72 = 0$ by Ferrari method. [05]

(c) Find the Fourier series to represent e^{ax} in the interval $-\pi \leq x \leq \pi$. [04]

SECTION-II

- Q-4 (a) State Divergence theorem. [02]
 (b) Show that the frequency of free vibrations in a closed electrical circuit with inductance L and capacity in series is $\frac{30}{\pi\sqrt{LC}}$ cycles/minute [05]

- Q-5 (a) A Particle moves along the curve $x = t^3 + 1, y = t^2, z = 2t + 5$, where t is the time. Find the components of its velocity and acceleration at time $t=1$ in the direction $2\hat{i} + 3\hat{j} + 6\hat{k}$. [05]
 (b) Find the directional derivative of $f(x, y, z) = x^2y^2z^2$ at the point $(1, 1, -1)$ in the direction of tangent to the curve $x = e^t, y = 2\sin t + 1, z = t - \cos t$ at $t=0$. [05]
 (c) if $\vec{a} = 2x^2\vec{i} - 3yz\vec{j} + xz^2\vec{k}$ and $\phi(x, y, z) = 2z^2 - x^3y$ find $\vec{a} \times \nabla \phi$ at the point $(1, -1, 1)$. [04]

OR

- Q-5 (a) A fluid motion is given by $\vec{v} = yz\vec{i} + xz\vec{j} + xy\vec{k}$ is the motion irrotational? If so, find the a scalar function $u(x, y, z)$ such that $\vec{v} = \text{grad } u$. [05]
 (b) Find the directional derivative of $\text{div } \vec{u}$ at the point $(1, 2, 2)$ in the direction of the outer normal of the sphere $x^2 + y^2 + z^2 = 109$ for $\vec{u} = xy\vec{i} + xy^2\vec{j} + z^2\vec{k}$. [05]
 (c) Find the divergence and curl of the vector field $\vec{v}(x, y, z) = xy^2\vec{i} + 2x^2y\vec{j} - 3yz^2\vec{k}$ at the point $(1, -1, 1)$. [04]

- Q-6 (a) Verify Green's theorem in the plane for $\int_C (xy + y^2)dx + (x^2)dy$, where C is the closed curve of the region bounded by $y = x^2$ & $y = x$. [05]
 (b) Verify Stoke's theorem for $\vec{f} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$ taken round the rectangle bounded by $x = \pm a, y = 0$ & $y = b$. [05]
 (c) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = 3xy\vec{i} - 5z\vec{j} + 10x\vec{k}$ and C is the arc of the curve $\vec{r} = (t^2 + 1)\vec{i} + 2t^2\vec{j} + t^3\vec{k}$ where t varies from 1 to 2. [04]

OR

- Q-6 (a) Using Divergence theorem to evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = 4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}$ and S is the surface bounding the region $x^2 + y^2 = 4, z = 0$ and $z = 3$. [05]
 (b) State Green's theorem and using theorem to evaluate $\int_C (x^2 + xy)dx + (x^2 + y^2)dy$ where C is the square formed by lines $y = \pm 1, x = \pm 1$. [05]
 (c) Compute $\int_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = \frac{y\vec{i} - x\vec{j}}{x^2 + y^2}$ and C is the circle $x^2 + y^2 = 1$ traversed counter clockwise. [04]
